

Universidad Autónoma "Gabriel René Moreno"
Facultad de Ingeniería en Ciencias de la Computación y
Telecomunicaciones
"School of Engineering"



Programa de Maestría

Maestría en Inteligencia Artificial

Módulo 13: Practicing Data Science: Práctica Opcional - Databricks

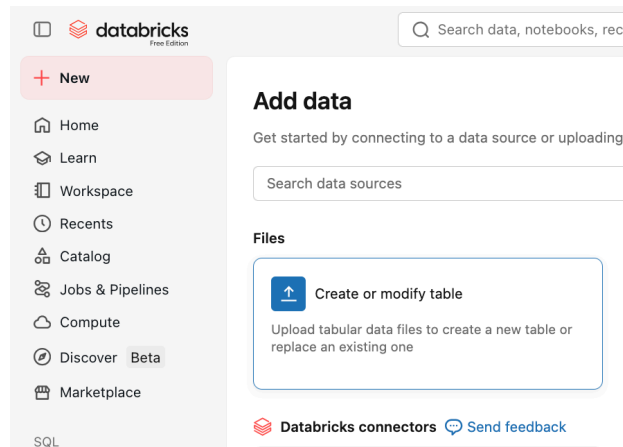
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Mayo 2026

Practica Opcional – Databricks

1. Descargar el dataset de Supermarket Sales desde:
<https://www.kaggle.com/datasets/faresashraf1001/supermarket-sales>
Se renombró las columnas para tener notación snake case, por ejemplo: “Tax 5%” a tax_five_percent.
2. En databricks, seleccionar “Create or modify table”

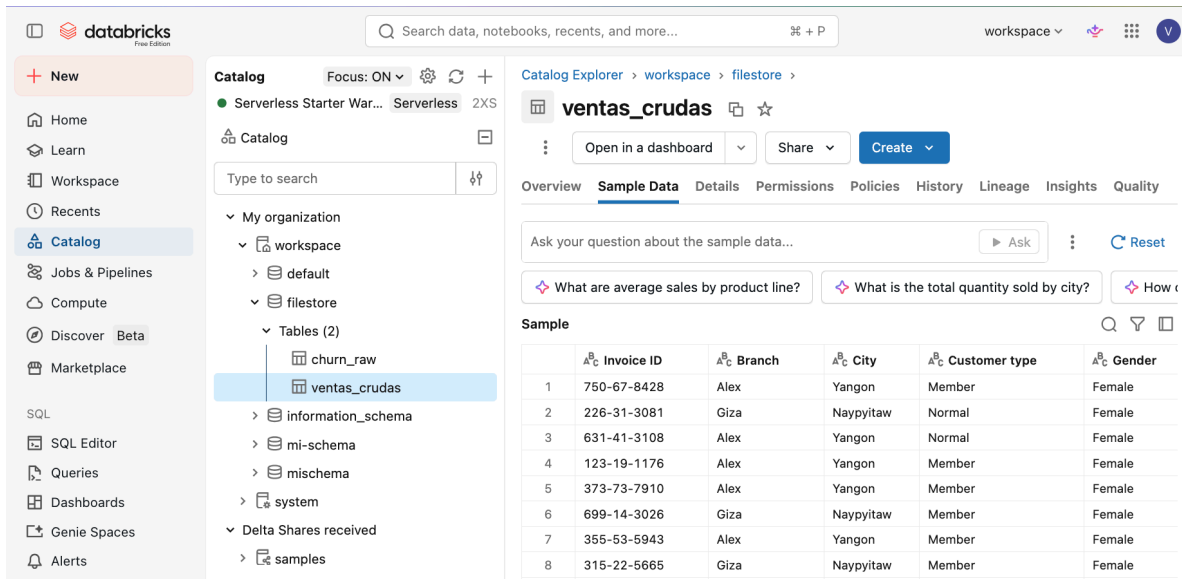


3. Seleccionar el dataset y nombrar la tabla como ventas_crudas, click en “Create table”

Invoice ID	Branch	City	Customer type	Gender
750-67-8428	Alex	Yangon	Member	Female
226-31-3081	Giza	Naypyitaw	Normal	Female
631-41-3108	Alex	Yangon	Normal	Female
123-19-1176	Alex	Yangon	Member	Female
373-73-7910	Alex	Yangon	Member	Female
699-14-3026	Giza	Naypyitaw	Member	Female
355-53-5943	Alex	Yangon	Member	Female
315-22-5665	Giza	Naypyitaw	Member	Female
665-32-9167	Alex	Yangon	Member	Female
692-92-5582	Cairo	Mandalay	Member	Female
351-62-0822	Cairo	Mandalay	Member	Female
529-56-3974	Cairo	Mandalay	Member	Female

At the bottom, there are 'Cancel' and 'Create table' buttons.

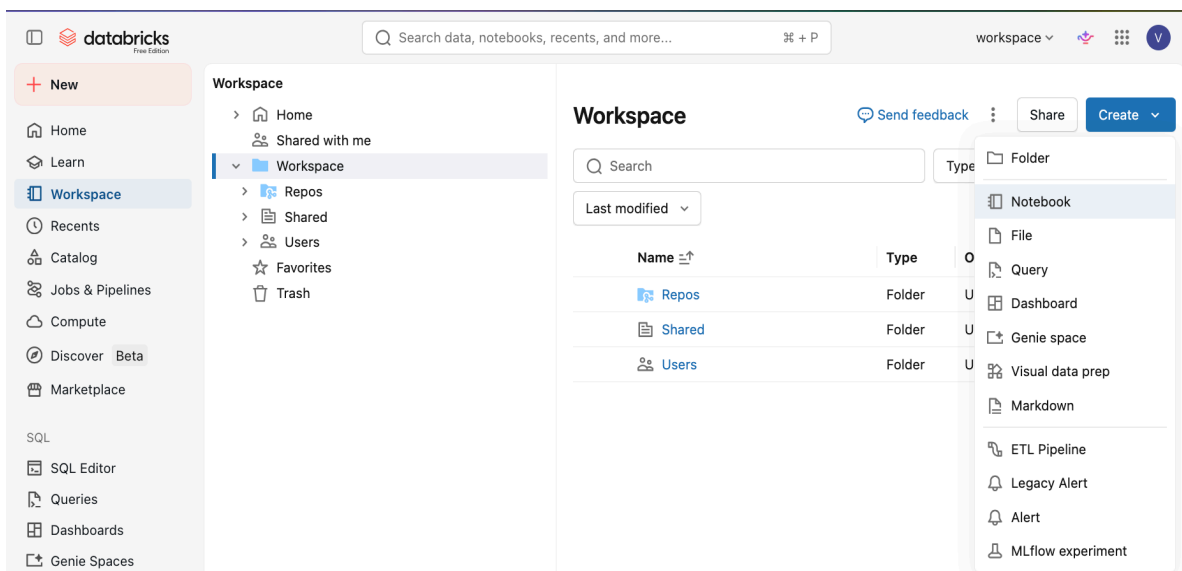
4. Verificar que se visualizan los datos de la tabla en “Sample Data”



The screenshot shows the Databricks Catalog Explorer interface. The left sidebar contains navigation options like Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, and Alerts. The main panel displays the 'ventas_crudas' table under the 'filestore' workspace. The 'Sample Data' tab is selected, showing a table with 8 rows of sample data. The table has columns: Invoice ID, Branch, City, Customer type, and Gender.

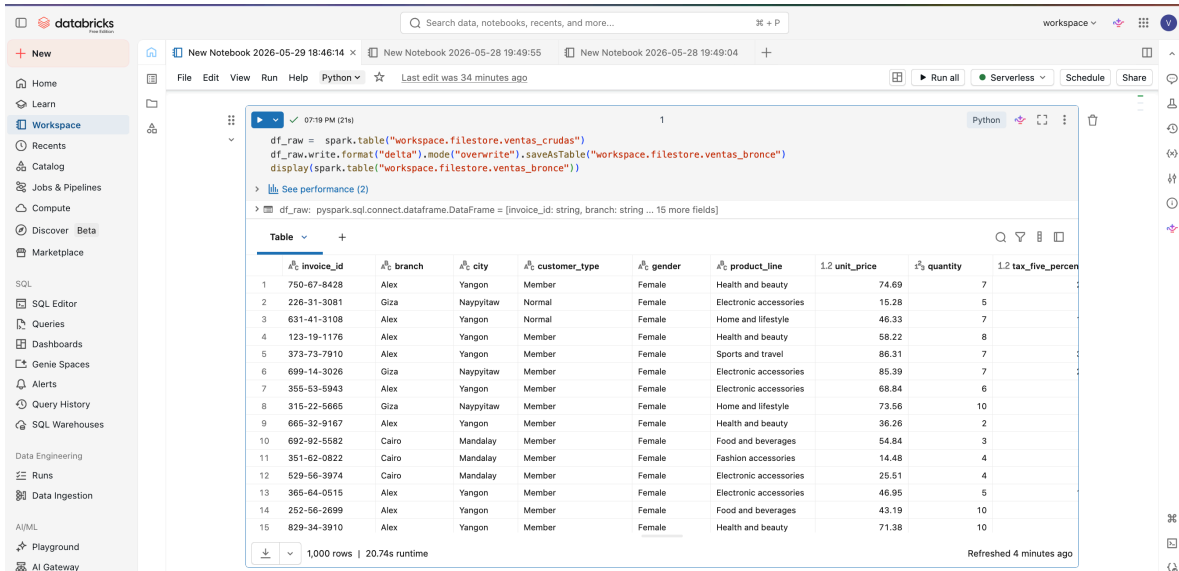
	Invoice ID	Branch	City	Customer type	Gender
1	750-67-8428	Alex	Yangon	Member	Female
2	226-31-3081	Giza	Naypyitaw	Normal	Female
3	631-41-3108	Alex	Yangon	Normal	Female
4	123-19-1176	Alex	Yangon	Member	Female
5	373-73-7910	Alex	Yangon	Member	Female
6	699-14-3026	Giza	Naypyitaw	Member	Female
7	355-53-5943	Alex	Yangon	Member	Female
8	315-22-5665	Giza	Naypyitaw	Member	Female

5. En el Workspace crear un nuevo notebook:



The screenshot shows the Databricks Workspace interface. The left sidebar contains navigation options like Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, and Alerts. The main panel displays the 'Workspace' tab, showing a list of folders: Repos, Shared, and Users. A context menu is open, showing options to create a new Notebook, File, Query, Dashboard, Genie space, Visual data prep, Markdown, ETL Pipeline, Legacy Alert, Alert, or MLflow experiment.

6. Crear una tabla utilizando spark ventas_bronce:



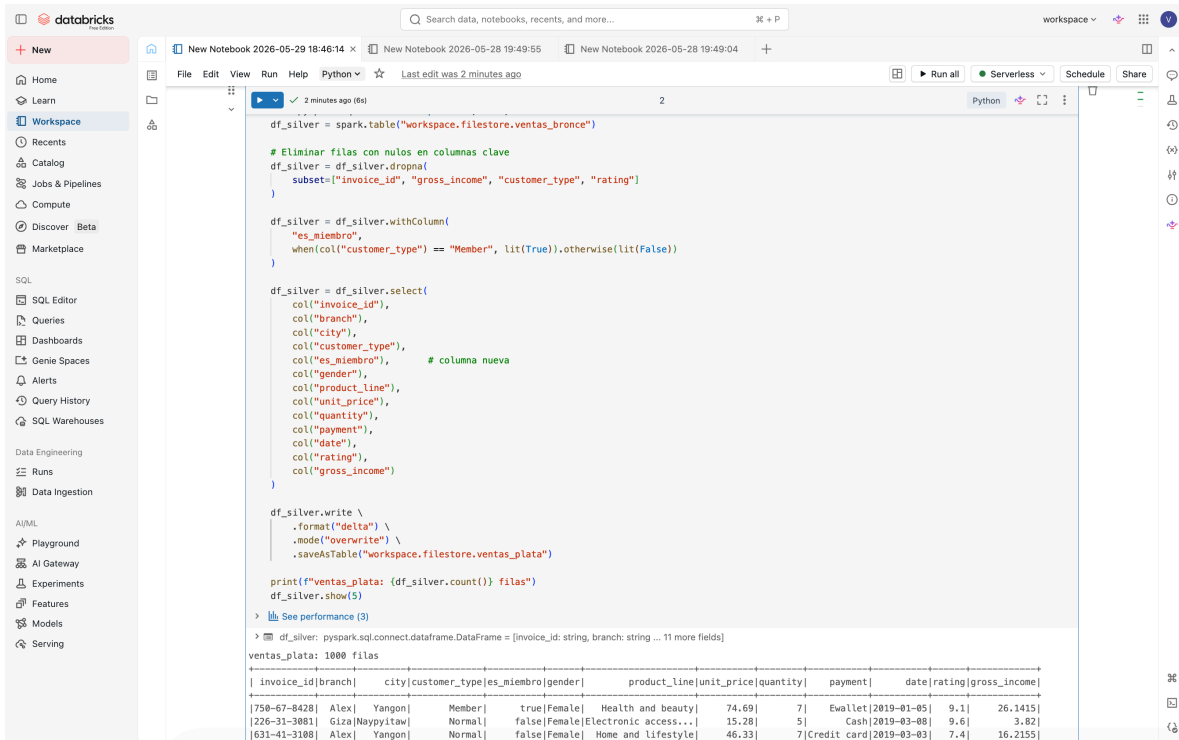
The screenshot shows a Databricks workspace with a notebook titled "New Notebook 2026-05-29 18:46:14". The notebook contains the following Python code:

```
df_raw = spark.table("workspace.filestore.ventas_crudas")
df_raw.write.format("delta").mode("overwrite").saveAsTable("workspace.filestore.ventas_bronce")
display(spark.table("workspace.filestore.ventas_bronce"))
```

Below the code, a preview of the table is shown with 15 rows and 9 columns:

	invoice_id	branch	city	customer_type	gender	product_line	unit_price	quantity	tax_five_percent
1	750-67-8428	Alex	Yanong	Member	Female	Health and beauty	74.69	7	
2	226-31-3081	Giza	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	
3	631-41-3108	Alex	Yanong	Normal	Female	Home and lifestyle	46.33	7	
4	123-19-1176	Alex	Yanong	Member	Female	Health and beauty	58.22	8	
5	373-73-7910	Alex	Yanong	Member	Female	Sports and travel	86.31	7	
6	699-14-3026	Giza	Naypyitaw	Member	Female	Electronic accessories	85.39	7	
7	355-53-5943	Alex	Yanong	Member	Female	Electronic accessories	68.84	6	
8	315-22-5665	Giza	Naypyitaw	Member	Female	Home and lifestyle	73.56	10	
9	665-32-9167	Alex	Yanong	Member	Female	Health and beauty	36.26	2	
10	692-92-5582	Cairo	Mandalay	Member	Female	Food and beverages	54.84	3	
11	351-62-0822	Cairo	Mandalay	Member	Female	Fashion accessories	14.48	4	
12	529-56-3974	Cairo	Mandalay	Member	Female	Electronic accessories	25.51	4	
13	365-64-0515	Alex	Yanong	Member	Female	Electronic accessories	46.95	5	
14	252-56-2699	Alex	Yanong	Member	Female	Food and beverages	43.19	10	
15	829-34-3910	Alex	Yanong	Member	Female	Health and beauty	71.38	10	

7. Crear una tabla , eliminar nulos, crear una nueva columna si es Miembro utilizando Member True o False, usando de identificador **Invoice ID**
8. Utilizar la variable dependente de ingresos **Gross Income**
9. Guardar en ventas_plata



The screenshot shows a Databricks workspace with a notebook titled "New Notebook 2026-05-29 18:46:14". The notebook contains the following Python code:

```
df_silver = spark.table("workspace.filestore.ventas_bronce")

# Eliminar filas con nulos en columnas clave
df_silver = df_silver.dropna(
    subset=["invoice_id", "gross_income", "customer_type", "rating"]
)

df_silver = df_silver.withColumn(
    "es_miembro",
    when(col("customer_type") == "Member", lit(True)).otherwise(lit(False))
)

df_silver = df_silver.select(
    col("invoice_id"),
    col("branch"),
    col("city"),
    col("customer_type"),
    col("es_miembro"), # columna nueva
    col("gender"),
    col("product_line"),
    col("unit_price"),
    col("quantity"),
    col("payment"),
    col("date"),
    col("rating"),
    col("gross_income")
)

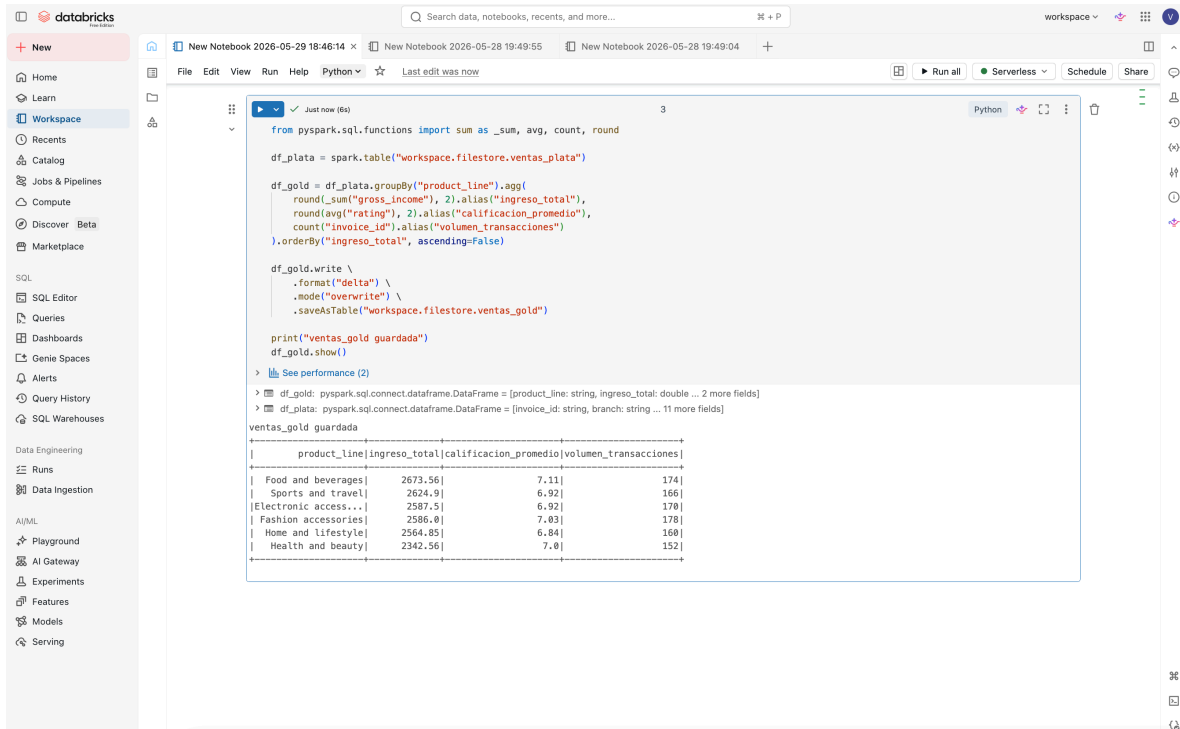
df_silver.write \
    .format("delta") \
    .mode("overwrite") \
    .saveAsTable("workspace.filestore.ventas_plata")

print(f"ventas_plata: {df_silver.count()} filas")
df_silver.show(5)
```

Below the code, a preview of the table is shown with 1000 rows and 11 columns:

invoice_id	branch	city	customer_type	es_miembro	gender	product_line	unit_price	quantity	payment	date	rating	gross_income
750-67-8428	Alex	Yanong	Member	true	Female	Health and beauty	74.69	7	Ewallet	2019-01-05	9.1	26.1415
226-31-3081	Giza	Naypyitaw	Normal	false	Female	Electronic access...	15.28	5	Cash	2019-03-08	9.6	3.82
631-41-3108	Alex	Yanong	Normal	false	Female	Home and Lifestyle	46.33	7	Credit card	2019-03-03	7.4	16.2155

10. Crear una tabla ventas_gold, suma de gross income, calcular la media de calificacion Rating, Contar las transacciones y lo denominan Volumen_transacciones



```
from pyspark.sql.functions import sum as _sum, avg, count, round

df_plata = spark.table("workspace.filestore.ventas_plata")

df_gold = df_plata.groupBy("product_line").agg(
    round(_sum("gross_income"), 2).alias("ingreso_total"),
    round(avg("rating"), 2).alias("calificacion_promedio"),
    count("invoice_id").alias("volumen_transacciones")
).orderBy("ingreso_total", ascending=False)

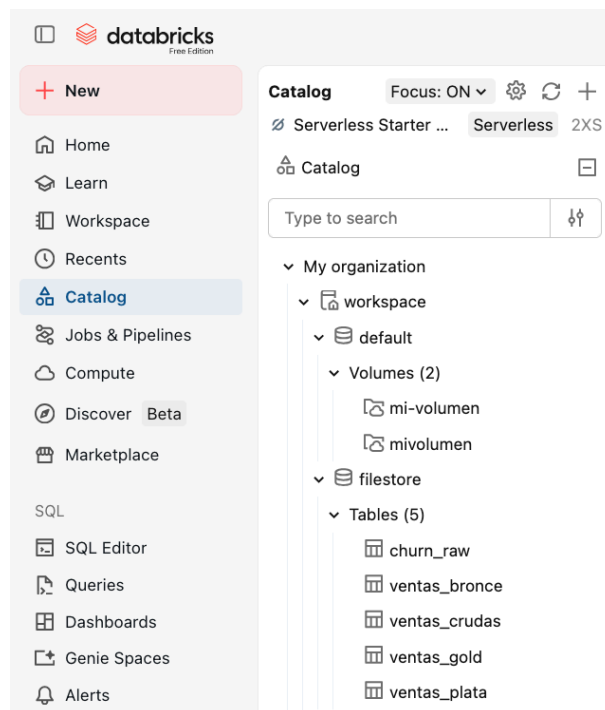
df_gold.write \
    .format("delta") \
    .mode("overwrite") \
    .saveAsTable("workspace.filestore.ventas_gold")

print("ventas_gold guardada")
df_gold.show()
```

ventas_gold guardada

product_line	ingreso_total	calificacion_promedio	volumen_transacciones
Food and beverages	2673.56	7.11	174
Sports and travel	2624.9	6.92	166
[Electronic access...]	2587.5	6.92	178
Fashion accessories	2586.0	7.83	178
Home and lifestyle	2564.85	6.84	160
Health and beauty	2342.56	7.0	152

Resultado de las tablas, bronce, plata y gold.



Catalog Focus: ON

Serverless Starter ... Serverless 2XS

Catalog

Type to search

- My organization
 - workspace
 - default
 - Volumes (2)
 - mi-volumen
 - mivolumen
 - filestore
 - Tables (5)
 - churn_raw
 - ventas_bronce
 - ventas_crudas
 - ventas_gold
 - ventas_plata

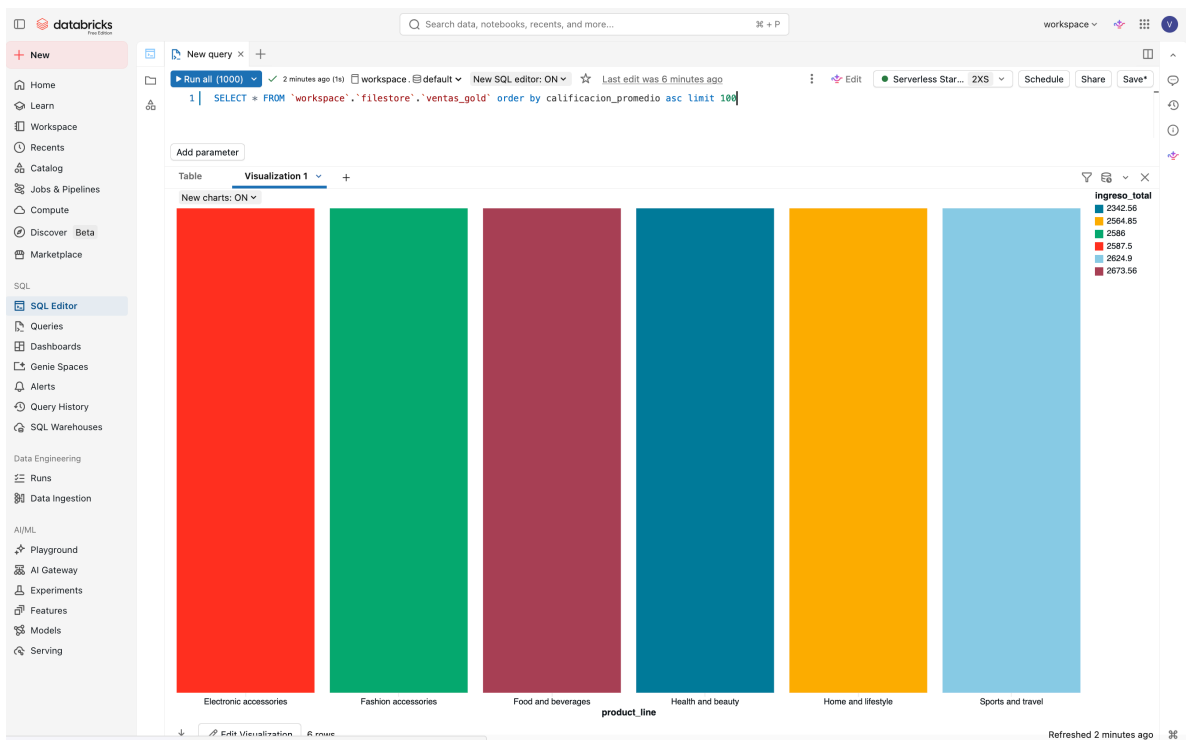
11. Crear un dashboard con un gráfico de barras con las líneas de producto e ingreso total

The screenshot shows the Databricks SQL Editor interface. The query editor contains the following SQL query:

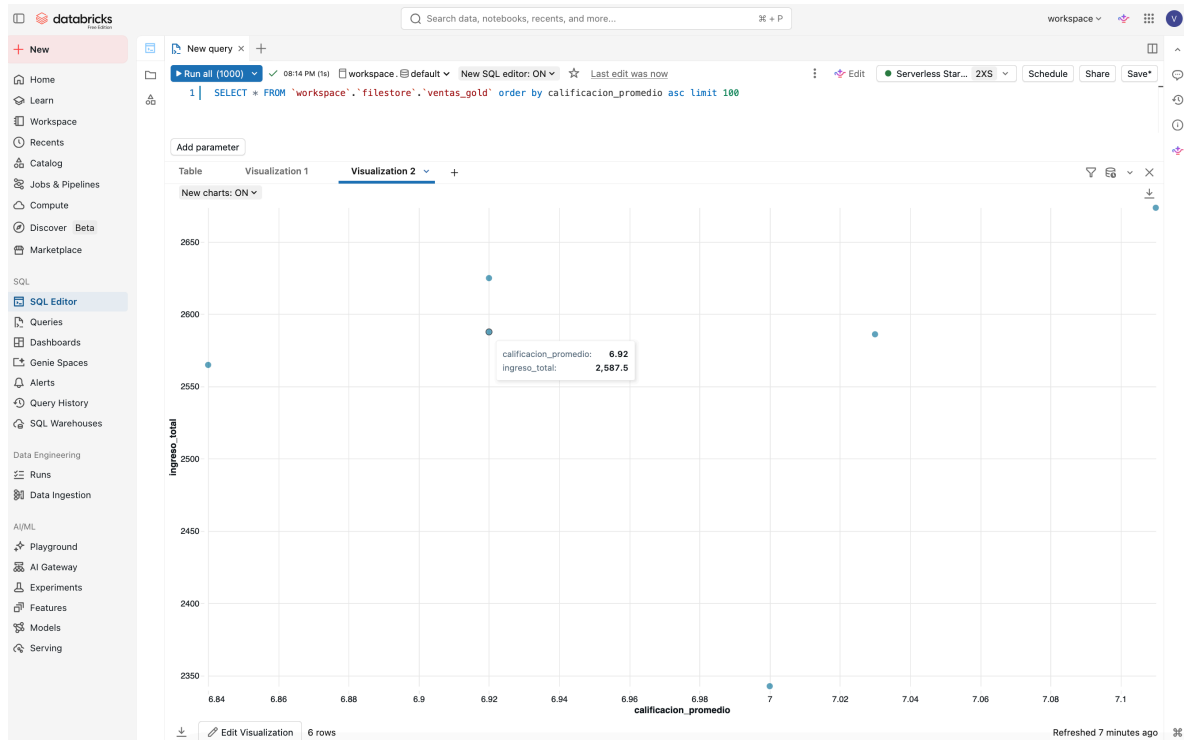
```
1 | SELECT * FROM `workspace`.`filestore`.`ventas_gold` order by calificacion_promedio asc limit 100
```

Below the query editor, the results are displayed in a table format:

	product_line	1.2 ingreso_total	1.2 calificacion_promedio	1.3 volumen_transacciones
1	Home and lifestyle	2564.85	6.84	160
2	Electronic accessories	2587.5	6.92	170
3	Sports and travel	2624.9	6.92	166
4	Health and beauty	2342.56	7	152
5	Fashion accessories	2586	7.03	178
6	Food and beverages	2673.56	7.11	174



12. Agregar un grafico Scatter, calificacion promedio y el ingreso total



13. Colocar un titulo

